

LAB	PRACTICALS
10	Implementation of Math Functions in SQL Part - A - Display the result of 5 multiply by 30. <code>SELECT 5*30 AS RESULT</code> - Find out the absolute value of -25, 25, -50 and 50. <code>SELECT ABS(-25) AS [ABS_-25], ABS(25) AS [ABS_25], ABS(-50) AS [ABS_-50], ABS(50) AS [ABS_50]</code> - Find smallest integer value that is greater than or equal to 25.2, 25.7 and -25.2. <code>SELECT CEILING(25.2) AS [CEIL_25.2], CEILING(25.7) AS [CEIL_25.7], CEILING(-25.2) AS [CEIL_-25.2]</code> - Find largest integer value that is smaller than or equal to 25.2, 25.7 and -25.2. <code>SELECT FLOOR(25.2) AS [FLOOR_25.2], FLOOR(25.7) AS [FLOOR_25.7], FLOOR(-25.2) AS [FLOOR_-25.2]</code> - Find out remainder of 5 divided 2 and 5 divided by 3. <code>SELECT 5%2 AS RES_1, 5%3 AS RES_2</code> Part – B - Find out value of 3 raised to 2nd power and 4 raised 3rd power. <code>SELECT POWER(3,2) AS POW_1, POWER(4,3) AS POW_2</code> - Find out the square root of 25, 30 and 50. <code>SELECT SQRT(25) AS [SQRT_25], SQRT(30) AS [SQRT_30], SQRT(50) AS [SQRT_50]</code> - Find out the square of 5, 15, and 25. <code>SELECT SQUARE(5) AS [SQ_5], SQUARE(15) AS [SQ_15], SQUARE(25) AS [SQ_25]</code> - Find out the value of PI. <code>SELECT PI() AS PI_VAL</code> - Find out round value of 157.732 for 2, 0 and -2 decimal points. <code>SELECT ROUND(157.732,2) AS R1, ROUND(157.732,0) AS R2, ROUND(157.732,-2) AS R3</code> Part – C - Find out exponential value of 2 and 3. <code>SELECT EXP(2) AS EXP_2, EXP(3) AS EXP_3</code>

	12. Find out logarithm having base b having value 10 of 5 and 100. <pre>SELECT LOG10(5) AS LOG_5, LOG10(100) AS LOG_100</pre> 13. Find sine, cosine and tangent of 3.1415. <pre>SELECT SIN(3.1415) AS [SIN_3.1415], COS(3.1415) AS [COS_3.1415], TAN(3.1415) AS [TAN_3.1415]</pre> 14. Find sign of -25, 0 and 25. <pre>SELECT SIGN(-25) AS S1, SIGN(0) AS S2, SIGN(25) AS S3</pre> 15. Generate random number using function. <pre>SELECT RAND() AS RANDOM_VALUE</pre>
11	Implementation of String Functions in SQL Part - A 1. Find the length of following. (I) NULL (II) 'hello' (III) Blank <pre>SELECT LEN(NULL) AS [LEN_1], LEN('hello') AS [LEN_2], LEN('') AS [LEN_3]</pre> 2. Display your name in lower & upper case. <pre>SELECT UPPER('Kavya') AS UPPERCASE, LOWER('KAVYA') AS LOWERCASE</pre> 3. Display first three characters of your name. <pre>SELECT LEFT('Kavya',3) AS FIRST_3</pre> 4. Display 3rd to 10th character of your name. <pre>SELECT SUBSTRING('Vishnu Ramanujan',3,10) AS [SUBSTRING_NAME]</pre> 5. Write a query to convert 'abc123efg' to 'abcXYZefg' & 'abcabcabc' to 'ab5ab5ab5' using REPLACE. <pre>SELECT REPLACE('abc123efg','123','XYZ') AS REPLACED_TXT_1, REPLACE('abcabcabc','c','5') AS REPLACED_TXT_2</pre> 6. Write a query to display ASCII code for 'a','A','z','Z', 0, 9. <pre>SELECT ASCII('a') as ASCII_a, ASCII('A') as ASCII_A, ASCII('z') as ASCII_z, ASCII('Z') as ASCII_Z, ASCII('0') as ASCII_0, -- IF WE WRITE ASCII(0) IT WILL IMPLICITLY CONVERTS IT TO CHAR ASCII('9') as ASCII_9</pre> 7. Write a query to display character based on number 97, 65,122,90,48,57. <pre>SELECT CHAR(97) AS CHAR_97, CHAR(65) AS CHAR_65, CHAR(122) AS CHAR_122,</pre>

CHAR(90) AS CHAR_90,
CHAR(48) AS CHAR_48,
CHAR(57) AS CHAR_57

Part – B

8. Write a query to remove spaces from left of a given string 'hello world ' .
`SELECT LTRIM(' hello world ') AS TRIM_LEFT`
9. Write a query to remove spaces from right of a given string ' hello world ' .
`SELECT RTRIM(' hello world ') AS TRIM_RIGHT`
10. Write a query to display first 4 & Last 5 characters of 'SQL Server' .
`SELECT LEFT('SQL Server',4) AS FIRST_4,
RIGHT('SQL Server',5) AS LAST_5`
11. Write a query to convert a string '1234.56' to number (Use CAST()).
`SELECT CAST('1234.56' AS DECIMAL(10,2)) AS RESULT;`
12. Write a query to convert a float 10.58 to integer (Use CONVERT()).
`SELECT CONVERT(INT, 10.58) AS RESULT;`
13. Put 10 space before your name using function.
`SELECT SPACE(10) + 'Kavya' AS Spaced_name`
14. Combine two strings (Your name & Surname) using + sign as well as CONCAT ().
`SELECT 'Kavya ' + 'Vyas' AS Concatenated_String
SELECT CONCAT('Kavya ','Vyas') AS Concatenated_String`

Part – C

15. Find reverse of "Darshan".
`SELECT REVERSE('Darshan') AS Reversed_string`
16. Repeat your name 3 times.
`SELECT REPLICATE('Kavya',3) AS Replicated_string`
17. Write a query to display first 3 & Last 3 characters of 'your name' .
`SELECT LEFT('Kavya',3) AS FIRST_3,
RIGHT('Kavya',3) AS LAST_3`
18. From Data, returns the first non-null value in a list. (Use COALESCE())
`SELECT COALESCE(NULL, NULL, 'Hello', 'World') AS RESULT;`
19. Tests whether the expression is numeric. (Use ISNUMERIC())
`SELECT ISNUMERIC('19.57') AS RESULT --STRING IS CONVERTED TO NUMBER
SELECT ISNUMERIC('KAVYA') AS RESULT`
20. Search for "t" in string "Customer", and return its position. (Use CHARINDEX())
`SELECT CHARINDEX('t','CUSTOMER',1) AS CHARINDEX_CUSTOMER`

12 Implementation of DATE Functions in SQL
Part - A

1. Write a query to display the current date & time. Label the column Today_Date.
`SELECT GETDATE() AS Today_Date`

2. Write a query to find new date after 365 day with reference to today.
`SELECT DATEADD(day,365,GETDATE()) AS NEW_DATE_365`

3. Display the current date in a format that appears as may 5 1994 12:00AM.
`SELECT CONVERT(VARCHAR, GETDATE(), 100) AS RESULT;`

4. Display the current date in a format that appears as 03 Jan 1995.
`SELECT FORMAT(GETDATE(), 'dd MMM yyyy') AS FormattedDate;`

5. Display the current date in a format that appears as Jan 04, 96
`SELECT FORMAT(GETDATE(), 'MMM dd, yy') AS FormattedDate;`

Part – B

6. Write a query to find out total number of months between 31-Dec-08 and 31-Mar-09.
`SELECT DATEDIFF(MONTH, '2008-12-31', '2009-03-31') AS TOTAL_MONTHS;`

7. Write a query to find out total number of years between 25-Jan-12 and 14-Sep-10.
`SELECT DATEDIFF(YEAR, '2010-09-14', '2012-01-25') AS TOTAL_YEARS;`

8. Write a query to find out total number of hours between 25-Jan-12 7:00 and 26-Jan-12 10:30.
`SELECT DATEDIFF(HOUR, '2012-01-25 07:00', '2012-01-26 10:30') AS TOTAL_HOURS;`

9. Write a query to extract Day, Month, Year from given date 12-May-16.
`SELECT DAY('12-May-16') AS [DAY],
MONTH('12-May-16') AS [MONTH],
YEAR('12-May-16') AS [YEAR]`

10. Write a query that adds 5 years to current date.
`SELECT DATEADD(YEAR,5,GETDATE()) AS [5_YEARS_LATER]`

Part – C

11. Write a query to subtract 2 months from current date.
`SELECT DATEADD(MONTH, -2, GETDATE()) AS RESULT;`

12. Extract month from current date using datename () and datepart () function.
`SELECT DATENAME(MONTH,GETDATE()) AS DATENAME_CURRENTDATE,
DATEPART(MONTH,GETDATE()) AS DATEPART_CURRENTDATE`

13. Write a query to find out last date of current month.
`SELECT EOMONTH(GETDATE()) AS LAST_DATE`

14. Write a query to display date & time after 30 days from today.
`SELECT DATEADD(DAY,30,GETDATE()) AS [30_DAYS_LATER]`

15. Calculate your age in years and months.
`SELECT DATEDIFF(YEAR,'2006-11-08',GETDATE()) AS AGE_YEARS
SELECT DATEDIFF(MONTH,'2006-11-08',GETDATE()) AS AGE_MONTHS`